

Acoustics Lab Three

Report

3/13/2025

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Abstract:

This lab uses steel wire under tension to determine the speed of a pulse. Using this you can determine the mass per unit length of the wire. Experiment is derived from Dr. Slaton and a former UCA student.

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I. Introduction:

The objective of this lab is to determine the speed of a pulse on a taut wire and using that information determine its mass per unit length. Make sure that the methods support the equation for the wave speed on a stretched wire using the tension and mass per unit length.

Equipment List & Setup:

Item	Quantity
Guitar strings	2
pickup	1
sonometer	1
Kg masses and hanger	assortment
Triple beam balance	1
digital oscilloscope	1
Stand, 90 degree connector and 3-finger clamp	1
Wire cutters & meter stick	1



II. Theory:

A. Background:

After deriving the equation for the speed of a wave on a taut string or wire of uniform mass per unit length, we find:

$$\sqrt{T/\rho_L}$$

where T is the tension in the string in N and ρ_L is the mass per unit length in kg/m. The wire will be plucked with a finger nail or guitar pick. The resulting pulse will propagate down the wire with speed, v , in m/s. The taut wire can be plucked to generate a pulse that will travel down the wire to be detected by the electric guitar pickup coil. The pickup coil consists of tiny magnets (the metal cylinders) wrapped in many many coils of wire. A moving metal wire that disrupts the magnetic field lines will change the flux through the coil. Faraday's Law states that a change in magnetic flux results in an induced voltage in the coil which we can detect with the oscilloscope.

III. Experimental Procedure:

During the experiment, the potential difference generated when the pulse transverses the region of magnetic field will come in handy to determine the speed of the pulse on the wire for a given tension. When the pulse hits a bridge, it is inverted upon reflection and propagates back. This will cause a corresponding negative voltage generated in the coil which will be visible on the oscilloscope. Therefore, if you know the distance between the center of the coil and the bridge that caused the reflection then you can use the oscilloscope to measure the time between the pulse and its echo. The ratio of the total distance traveled and the total time gives the speed of the pulse.

Our goal is to verify that Equation 1 is correct. If we plot our experimental data as tension vs. velocity squared then the theory says the slope should be, ρL , the mass per unit length of the wire. We can experimentally and independently measure the mass per unit length by measuring the mass of a known length of the wire. A comparison of these two independent measures of ρL , with uncertainty! This will allow us to determine if Equation 1 is correct.

To perform this experiment, 2 different wires will be used. These are 26- and 46-gauge guitar strings. The results for the strings will allow for comparison. Each string is a nickel-plated steel wire wrapped around a tin-platted hex shaped steel core wire.

IV. Results:

Note: I made the mistake of physically writing down experiment results on paper. This caused me to waste a fair amount of time copying it into .csv files when the data should have just been recorded in them while carrying out the experiment. Please learn from this mistake of mine.

The image shown below are the on-paper resulting measurements made:

$D\text{-string} \rightarrow 0.17\text{g per } 10\text{cm} \rightarrow 1.7\text{g per } 1\text{m}$
 $\pm 0.01\text{g}$
 $Low\ E\text{-string} \rightarrow 0.645\text{g per } 10\text{cm} \rightarrow 6.5\text{g per } 1\text{m}$
 $\pm 0.01\text{g}$

$V = \sqrt{\frac{T}{\mu}}$ $V = \frac{\lambda}{T}$
 $T = \text{tension}$
 $\mu = \text{mass per unit length}$

Acoustics Lab 3 Data Sheet

Length of strings $\rightarrow 0.877\text{m} \rightarrow l/2 = 0.439\text{m}$

String detail $\rightarrow$ strings are nickel-plated steel wire wrapped around a tin-plated hex shaped steel core wire.

Pickup to wall w L shape bracket $= 0.946\text{m}$ so $V = \frac{2(0.439)}{T_{avg}}$

D-String Trials: 1kg to 5kg tension, 1.7g per 1m, angle = $90^\circ \pm 0.5^\circ$

1kg trial	2kg trial	3kg trial	4kg trial	5kg trial
# time(ms)	# time(ms)	# time(ms)	# time(ms)	# time(ms)
1 10.6	1 8.3	1 6.7	1 5.5	1 4.8
2 11.6	2 8.4	2 6.8	2 5.5	2 4.8
3 11.6	3 8.5	3 6.7	3 5.5	3 4.9
4 11.3	4 8.5	4 6.7	4 5.6	4 4.8
5 11.7	5 8.4	5 6.5	5 5.6	5 4.8
Avg: 11.36ms	Avg: 8.42ms	Avg: 6.68ms	Avg: 5.54	Avg: 4.82
$V_{avg} \rightarrow \sqrt{\frac{T}{\mu}}$	$V_{avg} = 73.2\%$	$V_{avg} = 98.8\%$	$V_{avg} = 124.6\%$	$V_{avg} = 150.2\%$
$\mu = 1.7 \times 10^{-3} \text{kg/m}$	Diameter of string $\rightarrow 0.65\text{mm}$			
Tension(N)	8.2 N	16.4 N	24.5 N	32.7 N

Low e-string trial: 1kg to 5kg tension, 6.5g per 1m, angle = $90^\circ \pm 0.5^\circ$

1kg trial	2kg trial	3kg trial	4kg trial	5kg trial
# time(ms)	# time(ms)	# time(ms)	# time(ms)	# time(ms)
1 17.9	1 13.1	1 11.5	1 9.3	1 8.8
2 17.2	2 13.3	2 11.2	2 10.0	2 8.5
3 17.3	3 14.0	3 11.2	3 9.7	3 8.8
4 18.3	4 13.7	4 10.9	4 10.0	4 8.4
5 18.6	5 14.0	5 11.0	5 9.7	5 8.8
Avg: 17.86ms	Avg: 13.62ms	Avg: 11.16ms	Avg: 9.74ms	Avg: 8.66ms
$V_{avg} \rightarrow \sqrt{\frac{T}{\mu}}$	$V_{avg} = 46.6\%$	$V_{avg} = 61.1\%$	$V_{avg} = 74.6\%$	$V_{avg} = 85.4\%$
$\mu = 6.5 \times 10^{-3} \text{kg/m}$	Diameter of string $\rightarrow 1.17\text{mm}$			
Tension(N)	8.2 N	16.4 N	24.5 N	32.7 N

A. The Initial Numbers:

My lab partner and I set the pickup roughly at the middle of the string. We measured the distance from the pickup to the wall with the L-shaped bracket to be 0.416m. The total length of the string being plucked was measured at 0.877m which means we were off center by about 23mm. There is also the vertical distance between the wire and the pickup which was measured at 6mm.

For the D-String I cut and weighed a 10cm portion of the string. After converting the measurement, I found that the D-String has a mass per unit length of:

$$\rho_{LD} = (1.7 \pm 0.1) \times 10^{-3} \frac{kg}{m}.$$

The diameter was measured at 0.65mm.

Using the same method I found the mass per unit length of the Low E-String to be:

$$\rho_{LE} = (6.5 \pm 0.1) \times 10^{-3} \frac{kg}{m}.$$

The diameter of this string was 1.17mm

Some math needed to be done to find the tension force acting on the string. This was done by using the law of sines to find angles necessary for torque equations. The torque equations were used to perform a statics problem that would give the force acting on the string in equilibrium. After completing my calculations, I determined that the tension on the strings is the same for both. These tension forces for each weight added are below:

- 1 kg $\rightarrow$ 8.20N
- 2 kg $\rightarrow$ 16.4N
- 3 kg $\rightarrow$ 24.5N
- 4 kg $\rightarrow$ 32.7N
- 5 kg $\rightarrow$ 40.9N

B. D-String Trials:

The D-String performed well during experimentation. We took 5 time measurements in milliseconds for each 1kg weight that was added. Below is a table containing each measured time for each trial as well as the average time. To note this is the time between the pulse echoes seen on the oscilloscope. This is the time between the wave first passing the pickup, bouncing off the wall, and then passing by the pickup again.

Measurement	Time(ms)-1kg	Time(ms)-2kg	Time(ms)-3kg	Time(ms)-4kg	Time(ms)-5kg
1	10.6	8.3	6.7	5.5	4.8
2	11.6	8.4	6.8	5.5	4.8
3	11.6	8.5	6.7	5.5	4.9
4	11.3	8.5	6.7	5.6	4.8
5	11.7	8.4	6.5	5.6	4.8
Avg. Time	11.36	8.42	6.68	5.54	4.82

These average times allowed us to calculate the wave speed using distance over time. The table below will show the velocity of the wave using distance over time versus the calculated velocity using equation 1. Calculations will be in meters per second:

Calculation	Velocity-1kg	Velocity-2kg	Velocity-3kg	Velocity-4kg	Velocity-5kg
Velocity (d/t)	73.2	98.8	124.6	150.2	172.6
Equ. 1 Velocity	69.5	98.2	120.0	138.7	155.1
Difference	3.7	0.6	4.6	11.5	17.5

C.Low E-String Trials:

The Low E-String performed okay. On the 1kg trials we did not get very reliable data due to the weight not pulling the string tight enough. There was a good amount of rattling which caused our data to not be as clear as the D-String trials. The process is all the same and the results are below:

Measurement	Time(ms)-1kg	Time(ms)-2kg	Time(ms)-3kg	Time(ms)-4kg	Time(ms)-5kg
1	17.9	13.1	11.5	9.3	8.8
2	17.2	13.3	11.2	10.0	8.5
3	17.3	14.0	11.2	9.7	8.8
4	18.3	13.7	10.9	10.0	8.4
5	18.6	14.0	11.0	9.7	8.8
Avg. Time	17.86	13.62	11.16	9.74	8.66

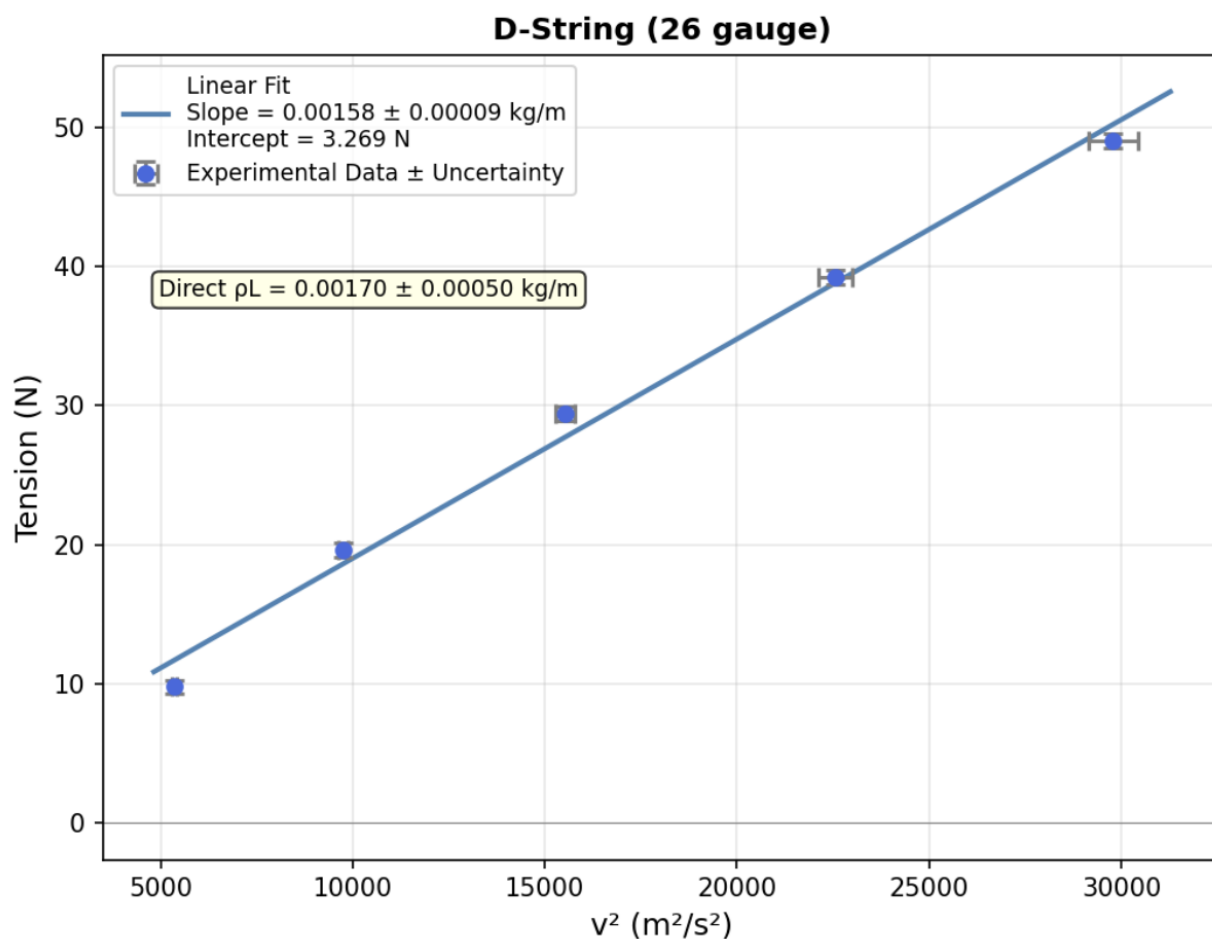
These average times allowed us to calculate the wave speed using distance over time. The table below will show the velocity of the wave using distance over time versus the calculated velocity using equation 1. You will notice that the differences between these velocity calculations are much greater than the D-String Trials. Calculations will be in meters per second:

Calculation	Velocity-1kg	Velocity-2kg	Velocity-3kg	Velocity-4kg	Velocity-5kg
Velocity (d/t)	46.6	61.1	74.6	85.4	96.1
Equ. 1 Velocity	35.5	50.2	61.4	70.9	79.3
Difference	11.1	10.9	13.2	15.5	16.8

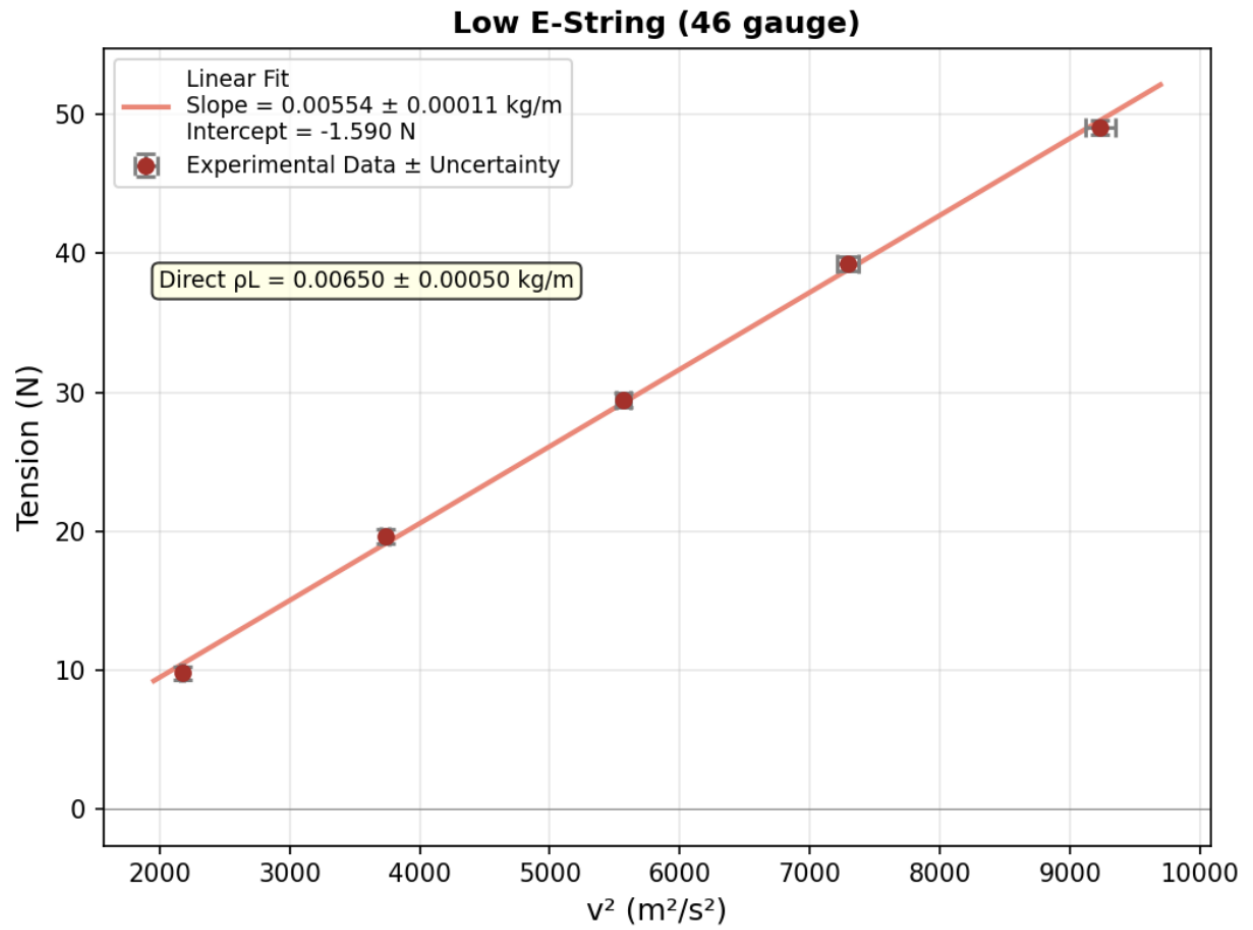
D. Data Plotting:

These plots were done by typing my data into .csv files. Then I told CoPilot what plots and information I wanted. Once it was done I got very good looking graphs that clearly display the data that we collected. The first graph displays the tension vs velocity squared for the D-String. The second graph displays the tension vs velocity squared for the Low E-String. The third graph is something I just wanted to visualize. It is a plot of the velocity vs hanging mass for both strings.

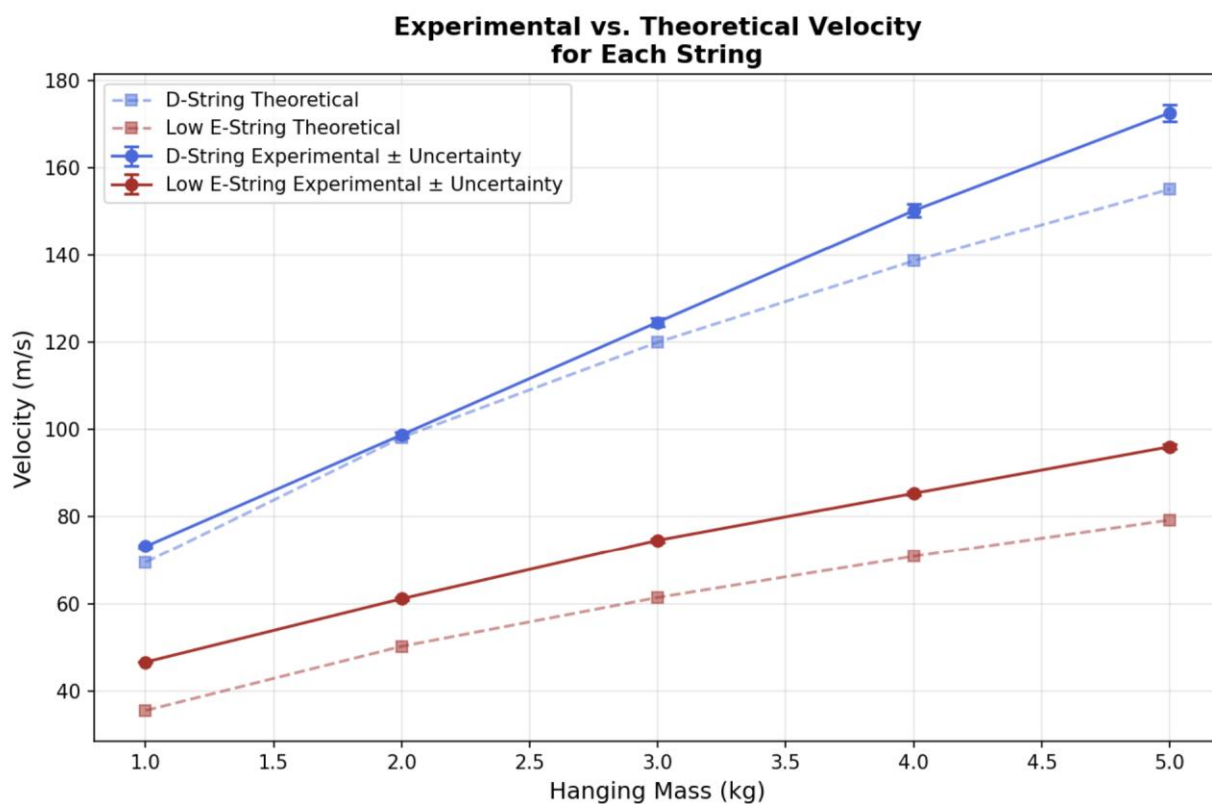
Graph 1:



Graph 2:



Graph 3:



The tables below are the linear fit results for both strings:

Measurement	D-String	Low E-String
Slope (kg/m)	$(157.6 \pm 8.8) \times 10^{-5}$	$(554.0 \pm 11.1) \times 10^{-5}$
Intercept (N)	3.279	-1.590
Estimated Mass per unit length (kg/m)	$(170.0 \pm 10) \times 10^{-5}$	$(650 \pm 10) \times 10^{-5}$
Agreement Check	AGREE	DISAGREE

V. Discussion:

A. Overall:

This lab overall was exciting since I got to donate some of my old guitar strings to the cause. They ended up working leagues better than the regular hardware store bought steel wire initially provided.

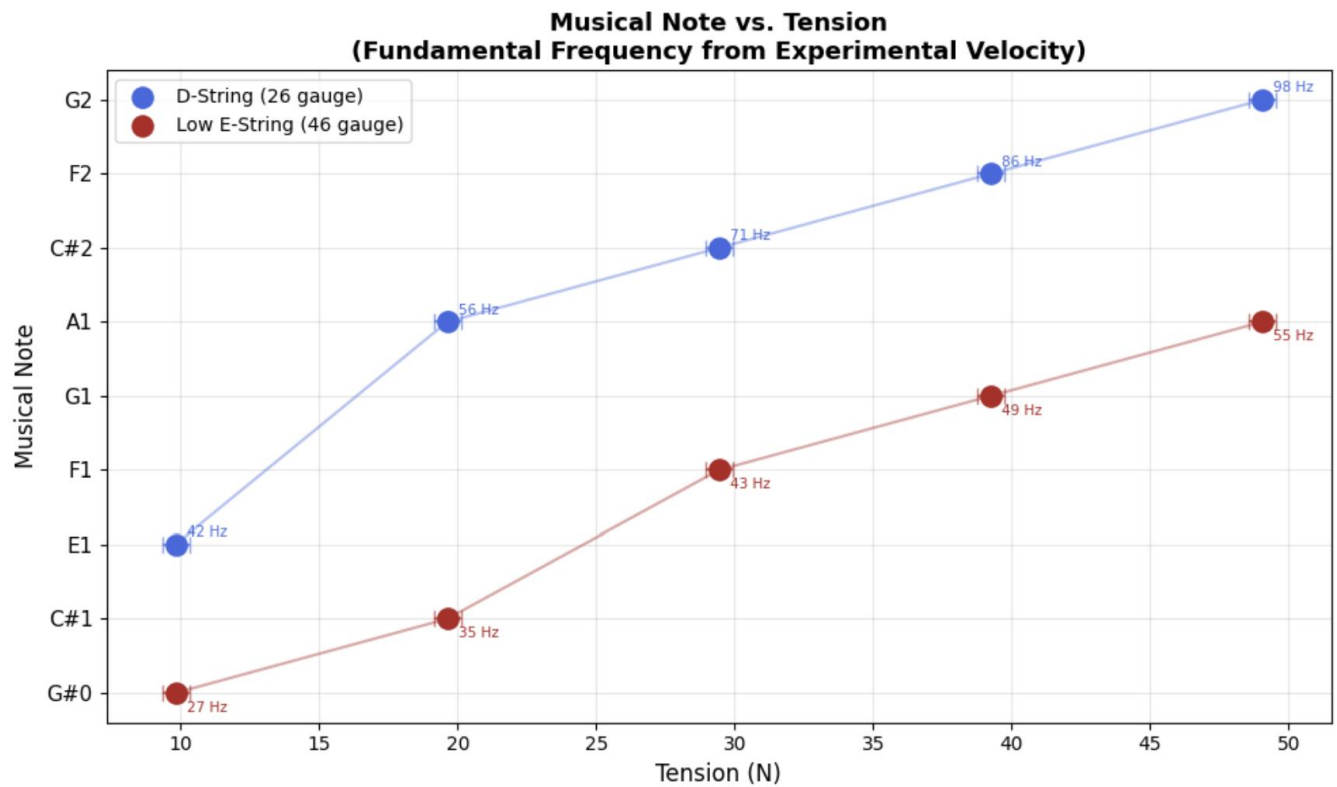
The experiment with the Low E-String should have been done with a shorter distance and heavier weights. That string can handle more tension due to its thickness. The data collected was bad which is why the experiment didn't pass the agreement check.

B. AI-Usage:

The usage of CoPilot inside of VSCode is great. Parts of this lab in my opinion need to be done with your own personal knowledge and reasoning skills. It is important to do this during experimentation as understanding where human error occurs or could happen plays a huge role in experimentation in all areas of study. Now once you've collected good sufficient data and know where faults might lie, I believe it is totally reasonable to use AI in creating graphs and summarizing data. Make sure to review its results as I had to correct a few things in mine.

VI. Conclusions:

I do not have a grand conclusion to this lab other than knowing where I messed up with data collection. In an ideal world it wouldn't be frustrating to erase collected to start over. During the creation of the plots I got curious as to what frequency and what note the string was producing. I created a plot of this as well which is displayed below:



VII. References:

Note: I need help with proper referencing

- [1] “Lab 3: Pulse on String”, Dr. Slaton Ph.D., 2026
- [2] CoPilot in VSCode, Creating plots and calculations.